

**Analyzing the use of quantitative geospatial analysis methods
for measuring passenger rail demand and the value of connectivity: Second Edition**

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Introduction

In March of 2026, I wrote a report for a research project in my *Quantitative Geospatial Analysis and Modeling (Geog361)* course at Oregon State University. The project focused on using quantitative spatial modeling to measure rail demand and the value of connectivity. This project is an extension of that research and further revisits the concepts, applies additional analysis, and covers a larger study area across different regions of the United States.

The passenger rail transportation market in the United States has undergone major changes, including a successful post-pandemic demand rebound and a successful expansion and enhancement of services (Congressional Research Service, 2025). However, *Amtrak*, the nation's primary passenger rail service, is subsidized by the federal government, which can involve a different business management model than the private sector. Funding and authorization depend on Congress. There is a concept in transportation economics known as "induced demand." In relation to highway traffic, this concept is applied when measuring changes in traffic demand because of highway expansion. Changes in vehicle traffic and highway demand are an example of demand for transportation being elastic and not fixed (Lee, n.d.). From FY2019 to FY2024, Congress appropriated a total of roughly \$30 billion for *Amtrak* (Congressional Research Service, n.d.). In contrast, from FY2019 to FY2024, Congress appropriated a total of roughly \$315 billion for the *Federal Highway Administration* (Federal Highway Administration, n.d.).

In addition to induced demand, the *Congressional Research Service* has acknowledged the value of connectivity for passenger rail (Congressional Research Service, 2025). Other studies have exhibited that new rail transportation can induce demand and expand entirely new markets, such as the increased real estate development and other land use changes indicating induced demand (Du, et. al., 2023). Despite this evidence, Congress seems unconvinced of the value of

connectivity in *Amtrak* expansion, and congressional budget appropriations for *Amtrak* remain at a fraction of the value of budget appropriations for the *Federal Highway Administration*.

This edition of the project was aimed at exploring and analyzing ridership data for some of *Amtrak*'s recently expanded services. This included entirely new routes that previously had no service, routes with pre-existing service that have recently undergone service enhancement, and routes undergoing service enhancement. By exploring and analyzing the ridership data and comparing entirely new services to expanded and pre-existing services, we can examine the applicability of quantitative geospatial analysis methods to support the concepts of induced demand and connectivity value. The original project's analysis was focused on the stations that are part of the newly expanded *Mardi Gras Service* from New Orleans, Louisiana to Mobile, Alabama. This new service has demonstrated success, and it has outperformed initial expectations (Southern Rail Commission, 2025). As such, ridership data analysis was limited to three states: Louisiana, Mississippi, and Alabama.

The new edition of this project also uses data for three other routes with established service: *Borealis*, *Pacific Surfliner*, and a section of the *Northeast Regional* in Virginia. These will be referred to as: *Borealis*, *Pacific Surfliner*, and *Virginia Southern*, respectively. In 2025, *Amtrak* added two daily round trip services to *Pacific Surfliner* (Amtrak Pacific Surfliner, 2025). In 2020, the Virginia Passenger Rail Authority was established to oversee implementation of further rail transportation expansion in Virginia, and when the current construction phase is completed in 2030, the state will fund 13 daily round trip services compared to the current 8 daily round trip services, those of which will include *Virginia Southern* (Amtrak, n.d.). *Borealis* began operations in 2024 and has been operating at capacity. *Borealis* shares right of way and stations with the *Empire Builder* and *Hiawatha Service* (Amtrak, 2026).

Borealis' termini are in St Paul, Minnesota and Chicago, Illinois. *Pacific Surfliner*'s termini are in San Luis Obispo, California and San Diego, California. *Virginia Southern* has a northern terminus in Alexandria, Virginia and two southern termini, Norfolk, Virginia and Newport News, Virginia, due to two separate branches. As such, the ridership data analysis for the second edition of the project was limited to relevant stops in California, Minnesota, Wisconsin, Illinois, and Virginia.

Figure 1: Initial Exploratory map

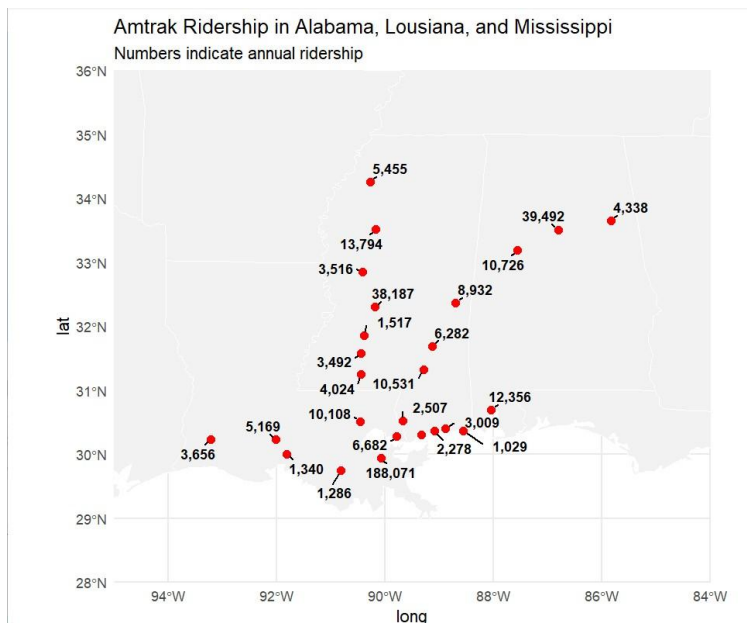


Fig. 1: This map was produced using R and RStudio, and it shows the distribution and annual ridership for Amtrak stations in Louisiana, Mississippi, and Alabama.

Methods

I used two primary datasets. The first one was annual *Amtrak* ridership data by station for 2025, which I acquired from the *Bureau of Transportation Statistics*. This will be referred to as “Dataset 1.” The second one was census data for a variety of counties relevant to the study, which I acquired from the *US Census Bureau*. This will be referred to as “Dataset 2.”

Dataset 1 initially included all annual ridership data for all stations in the entire United States. I downloaded *Dataset 1* from the website of the *Bureau of Transportation Statistics*. The data was downloaded into a Comma-separated values (.csv) file format. I manually copied the data for the three states of interest: Louisiana, Mississippi, and Alabama. Then, I manually pasted the data into a new .csv file in Microsoft Excel that was limited to the three states of interest. The file included columns for latitude (lat), longitude (lon), state (state), station name (station), and annual ridership (ridership) for each station. Lat, lon, and ridership were quantitative variables, with lat and lon being quantitative spatial variables. State and station were categorical variables. This produced the map in *Figure 1*.

Dataset 2 was a combination of datasets downloaded from the comparison table tool on the *Quick Facts* section of the website of the *US Census Bureau*. Due to limitations on size of comparison tables, I created and downloaded 4 separate tables for the first edition of the project. These tables were for the following categories: Counties of interest in Alabama, counties of interest in Mississippi, northernmost counties of interest in Louisiana, and southernmost counties of interest in Louisiana. Each table was downloaded as a .csv file, and I copied total population for each county. Then, I pasted the total population numbers for each county in a new .csv file in Microsoft Excel. The new .csv file for *Dataset 2* now had data from *Dataset 1* for ridership as well. *Dataset 2* had 5 columns. These included columns for the state of which the county is a part (state), the county's name (county), total annual ridership for all *Amtrak* stations within the county (ridership), total population of the county (population), and normalized ridership data for the county, measured in total annual ridership per 10,000 people (ride_10k). State and county are categorical variables, while ridership, population, and ride_10k are quantitative variables.

I used a similar method for acquiring data for the second edition of the project. The newly added data for *Dataset 2* was from counties of interest in Minnesota, Wisconsin, Illinois, California, and Virginia. The newly added data for *Dataset 1* was from stations along the routes of interest: *Borealis*, *Pacific Surfliner*, and *Virginia Southern*.

Dataset 1 and *Dataset 2* are my primary datasets that I downloaded from their primary sources, and I managed, filtered, and selected the datasets to create relevant and smaller datasets for my study area. In addition to the primary datasets, I also used the internal datasets within R and RStudio for state and county basemaps of the United States, which allowed for the joining of the basemap .csv files for Dataset 1 and Dataset 2.

Figure 2a: Dataset 1 (first edition)

	A	B	C	D	E	F
1	lat	lon	state	station	ridership	
2	33.649	-85.832	Alabama	Anniston,	4338	
3	33.5124	-86.807	Alabama	Birmingham,	39492	
4	30.6903	-88.038	Alabama	Mobile, Al	12356	
5	33.1923	-87.56	Alabama	Tuscaloosa,	10726	
6	30.5117	-90.464	Louisiana	Hammond,	10108	
7	30.2273	-92.015	Louisiana	Lafayette,	5169	
8	30.2381	-93.215	Louisiana	Lake Charles,	3656	
9	30.0085	-91.824	Louisiana	New Iberia,	1340	
10	29.9468	-90.08	Louisiana	New Orleans,	188071	
11	29.7469	-90.813	Louisiana	Schriever,	1286	
12	30.2782	-89.782	Louisiana	Slidell, La	6682	
13	30.3089	-89.334	Mississippi	Bay St. Louis,	2310	
14	30.3991	-88.891	Mississippi	Biloxi, Miss	3009	
15	31.5788	-90.442	Mississippi	Brookhaven,	3492	
16	33.5177	-90.176	Mississippi	Greenwood,	13794	
17	30.3692	-89.095	Mississippi	Gulfport, Miss	2278	
18	31.3267	-89.286	Mississippi	Hattiesburg,	10531	
19	31.8629	-90.394	Mississippi	Hazlehurst,	1517	
20	32.3006	-90.191	Mississippi	Jackson,	38187	
21	31.6924	-89.128	Mississippi	Laurel, Miss	6282	
22	34.2582	-90.275	Mississippi	Marks, Miss	5455	
23	31.2493	-90.453	Mississippi	McComb,	4024	
24	32.3638	-88.696	Mississippi	Meridian,	8932	
25	30.3678	-88.559	Mississippi	Pascagoula,	1029	
26	30.5247	-89.68	Mississippi	Picayune,	2507	
27	32.8481	-90.415	Mississippi	Yazoo City,	3516	

Fig. 2a: Dataset 1 includes ridership data and spatial information about each station of interest in the first edition of the project. Figure is a screenshot of the .csv file in Microsoft Excel.

Figure 2b: Additional data from Dataset 1 for California (second edition)

	A	B	C	D	E	F
1	lat	lon	state	station	ridership	
2	33.80426	-117.884819	California	Anaheim, California	267870	
3	38.018085	-121.815132	California	Antioch-Pittsburg, California	35667	
4	38.904182	-121.082031	California	Auburn, California	9348	
5	35.37138	-119.022461	California	Bakersfield, California	376353	
6	34.9048	-117.025352	California	Barstow, California	3300	
7	37.867603	-122.299995	California	Berkeley, California	99353	
8	34.20581553	-118.3492187	California	Burbank Airport, California	67923	
9	34.193832	-118.355911	California	Burbank, California	10852	
10	34.216469	-119.033112	California	Camarillo, California	52904	
11	34.396004	-119.521317	California	Carpenteria, California	27105	
12	34.257118	-118.598732	California	Chatsworth, California	58788	
13	39.723309	-121.844719	California	Chico, California	9749	
14	39.099457	-120.952705	California	Colfax, California	6615	
15	36.10083915	-119.5574567	California	Col. Allensworth Park, California	1651	
16	36.098194	-119.556503	California	Corcoran, California	21981	
17	38.544113	-121.737068	California	Davis, California	275354	
18	41.211174	-122.269974	California	Dunsmuir, California	3794	
19	37.840679	-122.29068	California	Emeryville, California	444451	
20	38.286164	-121.9703517	California	Fairfield-Vacaville, California	74502	
21	37.558197	-122.066378	California	Fremont, California	32781	
22	36.738628	-119.782249	California	Fresno, California	302753	
23	33.868946	-117.922836	California	Fullerton, California	240141	
24	34.123638	-118.257957	California	Glendale, California	53489	
25	34.437557	-119.841171	California	Goleta, California	86988	
26	35.122066	-120.628105	California	Grover Beach, California	14563	
27	34.962505	-120.572945	California	Guadalupe-Santa Maria, California	12287	
28	36.326214	-119.65078	California	Hanford, California	121136	
29	37.666859	-122.097786	California	Hayward, California	36948	
30	33.674767	-117.759193	California	Irvine, California	244373	
31	38.13354	-121.271594	California	Lodi, California	2769	
32	34.643879	-120.456833	California	Lompoc-Surf, California	9025	
33	34.056297	-118.236534	California	Los Angeles, California	1066614	
34	36.974518	-120.018944	California	Madera, California	25476	
35	38.019733	-122.136353	California	Martinez, California	267797	
36	37.307041	-120.475891	California	Merced, California	135210	
37	37.66814	-120.911598	California	Modesto, California	105596	
38	34.285194	-118.876953	California	Moorpark, California	37744	
39	34.840279	-114.604874	California	Needles, California	7395	
40	34.23066679	-118.5454514	California	Northridge, California	24364	
41	37.75354099	-122.157674	California	Oakland Coliseum, California	24285	
42	37.75370471	-122.2715055	California	Oakland, California	246499	
43	33.192581	-117.378624	California	Oceanside, California	211534	
44	34.061707	-117.647858	California	Ontario, California	4884	

Fig. 2b: This screenshot of Dataset 1 is a .csv file that shows stations in California for the second edition.

Figure 2c: Additional data from Dataset 1 for California and Illinois (second edition)

	A	B	C	D	E	F
45	34.198833	-119.174966	California	Oxnard, California	92195	
46	33.826698	-116.544914	California	Palm Springs, California	1688	
47	35.622925	-120.686935	California	Paso Robles, California	9510	
48	34.053995	-117.750633	California	Pomona, California	1434	
49	40.582832	-122.391922	California	Redding, California	8505	
50	37.936085	-122.35305	California	Richmond, California	202257	
51	33.976521	-117.369621	California	Riverside, California	9468	
52	38.791546	-121.236275	California	Rocklin, California	8912	
53	38.750565	-121.284462	California	Roseville, California	37157	
54	38.58403	-121.49955	California	Sacramento, California	764100	
55	36.679363	-121.656189	California	Salinas, California	20833	
56	34.104362	-117.306931	California	San Bernardino, California	9576	
57	33.419159	-117.619011	California	San Clemente Pier, California	16486	
58	32.716164	-117.168564	California	San Diego, California	489962	
59	32.75545496	-117.1999881	California	San Diego-Old Town, California	308805	
60	37.33107	-121.901207	California	San Jose, California	156081	
61	33.501484	-117.863261	California	San Juan Capistrano, California	162211	
62	35.277416	-120.653946	California	San Luis Obispo, California	91556	
63	33.753166	-117.855042	California	Santa Ana, California	109990	
64	34.413925	-119.69072	California	Santa Barbara, California	339569	
65	37.395944	-121.946411	California	Santa Clara (Great America), California	83287	
66	37.35199728	-121.930202	California	Santa Clara (University), California	40932	
67	34.269947	-118.693718	California	Simi Valley, California	40003	
68	32.992054	-117.270073	California	Solana Beach, California	202986	
69	37.956668	-121.278989	California	Stockton (Downtown), California	5094	
70	37.946011	-121.284271	California	Stockton (San Joaquin St.), California	274640	
71	38.243465	-122.039543	California	Suisun-Fairfield, California	65360	
72	39.327915	-120.182571	California	Truckee, California	16496	
73	37.530312	-120.798798	California	Turlock-Denair, California	31978	
74	34.210979	-118.447322	California	Van Nuys, California	67885	
75	34.276752	-119.296577	California	Ventura, California	79936	
76	34.536629	-117.29213	California	Victorville, California	7228	
77	35.594273	-119.331352	California	Wasco, California	23549	
78	38.904491	-90.134949	Illinois	Alton, Illinois	75548	
79	40.508347	-89.984039	Illinois	Bloomington/Normal, Illinois	230448	
80	37.724087	-89.21637	Illinois	Carbondale, Illinois	82122	
81	39.279327	-89.888916	Illinois	Carlinville, Illinois	9349	
82	38.527771	-89.136192	Illinois	Centralia, Illinois	17638	
83	40.116196	-89.240929	Illinois	Champaign-Urbana, Illinois	254996	
84	41.878731	-87.639168	Illinois	Chicago, Illinois	3175856	
85	38.011353	-89.240158	Illinois	Du Quoin, Illinois	6909	
86	41.092049	-88.428322	Illinois	Dwight, Illinois	4964	
87	39.119156	-88.545715	Illinois	Elmhurst, Illinois	22775	
88	40.94405	-90.364105	Illinois	Galesburg, Illinois	71850	

Fig. 2c: This screenshot of the .csv file of Dataset 1 shows more stations in California, as well as stations in Illinois for the second edition.

Figure 2d: Additional data from Dataset 1 for Illinois, Minnesota, and Virginia (second edition)

	A	B	C	D	E	F
89	40.762077	-87.994766	Illinois	Gilman, Illinois	2172	
90	42.074196	-87.805344	Illinois	Glenview, Illinois	57752	
91	41.562351	-87.66851	Illinois	Homewood, Illinois	46211	
92	41.524876	-88.07914	Illinois	Joliet, Illinois	69420	
93	41.119133	-87.865761	Illinois	Kankakee, Illinois	18244	
94	41.245625	-89.927422	Illinois	Kewanee, Illinois	13172	
95	41.815903	-87.86972	Illinois	La Grange Road, Illinois	9129	
96	40.147537	-89.363304	Illinois	Lincoln, Illinois	18239	
97	40.461132	-90.672279	Illinois	Macomb, Illinois	40454	
98	39.482422	-88.375526	Illinois	Mattoon, Illinois	34319	
99	41.550041	-89.117287	Illinois	Mendota, Illinois	18178	
100	41.779507	-88.14505	Illinois	Naperville, Illinois	34737	
101	41.66199036	-88.53835837	Illinois	Plano, Illinois	5725	
102	40.87944	-88.636497	Illinois	Pontiac, Illinois	14840	
103	41.386906	-89.461678	Illinois	Princeton, Illinois	29018	
104	39.957443	-91.366348	Illinois	Quincy, Illinois	29252	
105	40.31051202	-88.15972832	Illinois	Rantoul, Illinois	5865	
106	39.801586	-89.651672	Illinois	Springfield, Illinois	159999	
107	41.792164	-87.813179	Illinois	Summit, Illinois	19729	
108	46.819489	-95.845581	Minnesota	Detroit Lakes, Minnesota	4724	
109	44.564541	-92.54184	Minnesota	Red Wing, Minnesota	21666	
110	45.566891	-94.147789	Minnesota	St. Cloud, Minnesota	8247	
111	44.963329	-93.186325	Minnesota	St. Paul-Minneapolis, Minnesota	199613	
112	45.354588	-94.795113	Minnesota	Staples, Minnesota	6692	
113	44.044937	-91.641747	Minnesota	Winona, Minnesota	23673	
114	38.806244	-77.062645	Virginia	Alexandria, Virginia	372327	
115	37.759235	-77.481674	Virginia	Ashland, Virginia	42110	
116	38.797271	-77.29879	Virginia	Burke Centre, Virginia	14721	
117	38.032032	-78.492111	Virginia	Charlottesville, Virginia	199357	
118	37.811771	-79.836494	Virginia	Clifton Forge, Virginia	2178	
119	38.472923	-77.993042	Virginia	Culpeper, Virginia	21372	
120	36.583698	-79.384315	Virginia	Danville, Virginia	8324	
121	38.298557	-77.456932	Virginia	Fredericksburg, Virginia	114797	
122	38.705963	-77.221375	Virginia	Lorton (Auto Train), Virginia	265950	
123	37.407024	-79.157463	Virginia	Lynchburg, Virginia	79575	
124	38.750034	-77.473846	Virginia	Manassas, Virginia	52758	
125	37.019356	-76.45105	Virginia	Newport News, Virginia	109799	
126	36.85351544	-76.29112252	Virginia	Norfolk, Virginia	224375	
127	37.241619	-77.42897	Virginia	Petersburg, Virginia	60480	
128	38.522011	-77.293045	Virginia	Quantico, Virginia	20492	
129	37.53406	-77.42979	Virginia	Richmond - Main Street, Virginia	129928	
130	37.61425	-77.496582	Virginia	Richmond - Staples Mill, Virginia	506023	
131	37.2731571	-79.9423058	Virginia	Roanoke, Virginia	105926	
132	38.147781	-79.072723	Virginia	Staunton, Virginia	5519	

Fig. 2d: This screenshot of the .csv file of Dataset 1 shows more stations in Illinois, all stations in Minnesota, and stations in Virginia for the second edition.

Figure 2e: Additional data from Dataset 1 for Virginia and Wisconsin (second edition)

	A	B	C	D	E	F	G
133	37.275726	-76.708054	Virginia	Williamsburg, Virginia	74190		
134	38.657444	-77.248276	Virginia	Woodbridge, Virginia	37278		
135	43.340668	-89.012482	Wisconsin	Columbus, Wisconsin	23504		
136	43.8223168	-91.3569994	Wisconsin	La Crosse, Wisconsin	39963		
137	43.034016	-87.916481	Wisconsin	Milwaukee, Wisconsin	555161		
138	42.94174654	-87.92447572	Wisconsin	Milwaukee Airport, Wisconsin	123946		
139	43.546612	-89.469139	Wisconsin	Portage, Wisconsin	10162		
140	42.71879402	-87.90560196	Wisconsin	Sturtevant, Wisconsin	55779		
141	43.985725	-90.504166	Wisconsin	Tomah, Wisconsin	17998		
142	43.624786	-89.775078	Wisconsin	Wisconsin Dells, Wisconsin	24036		
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Fig. 2e: This screenshot of the .csv file of Dataset 1 shows 2 Virginia stations and all Wisconsin stations.

Figure 3a: Dataset 2 (first edition)

	A	B	C	D	E	F
1	state	county	ridership	population	ride_10k	
2	Alabama	Calhoun	4338	116,427	372.594	
3	Alabama	Jefferson	39492	664,744	594.0934	
4	Alabama	Mobile	12356	412,339	299.6564	
5	Alabama	Tuscaloos	10726	241,212	444.6711	
6	Louisiana	Tangipahoe	10108	139,823	722.914	
7	Louisiana	Lafayette	5169	254,241	203.311	
8	Louisiana	Calcasieu	3656	206,861	176.737	
9	Louisiana	Iberia	1340	67,540	198.4009	
10	Louisiana	Orleans	188071	362,701	5185.29	
11	Louisiana	Terrebonne	1286	103,864	123.8158	
12	Louisiana	St. Tammany	6682	277,615	240.693	
13	Mississippi	Hancock	2310	46,492	496.8597	
14	Mississippi	Harrison	5287	213,730	247.3682	
15	Mississippi	Jackson	1029	147,002	69.99905	
16						

Fig. 3a: This portion of Dataset 2 includes rows for each county of interest for the first edition of the project, with columns for county name and state name, plus population of each county, total annual ridership of all Amtrak stations in each county, and normalized ridership data of ridership per 10,000 people. Figure is a screenshot of the .csv file in Microsoft Excel.

Figure 3b: Dataset 2- Borealis (second edition)

	A	B	C	D	E
1	state	county	ridership	population	ride_10k
2	Illinois	Cook	3233608	5194625	6224.911
3	Wisconsin	Racine	55779	198919	2804.106
4	Wisconsin	Milwaukee	679107	924216	7347.925
5	Wisconsin	Columbia	57702	58132	9926.03
6	Wisconsin	Monroe	17998	46572	3864.554
7	Wisconsin	La Crosse	39963	121339	3293.5
8	Minnesota	Winona	23673	50523	4685.589
9	Minnesota	Goodhue	21666	48195	4495.487
10	Minnesota	Ramsey	199613	541623	3685.46

Fig. 3b: This .csv file of Dataset 2 includes rows for each county of interest for *Borealis* in the second edition of the project, with columns for county name and state name, plus population of each county, total annual ridership of all Amtrak stations in each county, and normalized ridership data of ridership per 10,000 people.

Figure 3c: Dataset 2- Pacific Surfliner (second edition)

	A	B	C	D	E
1	State	County	ridership	population	ride_10k
2	California	San Luis C	106119	282,367	3758.194
3	California	Santa Barl	475574	442,065	10758.01
4	California	Ventura	302782	830,851	3644.239
5	California	Los Angele	1339915	9,694,934	1382.077
6	California	Orange	981071	3,149,507	3114.999
7	California	San Diego	1213287	3,282,248	3696.512

Fig. 3c: This .csv file of Dataset 2 includes rows for each county of interest for *Pacific Surfliner* in the second edition of the project, with columns for county name and state name, plus population of each county, total annual ridership of all Amtrak stations in each county, and normalized ridership data of ridership per 10,000 people.

Figure 3d: Dataset 2- Virginia Southern

	A	B	C	D	E
1	State	County	ridership	population	ride_10k
2	Virginia	Alexandria	372327	160,662	23174.55
3	Virginia	Prince Will	57770	502,966	1148.587
4	Virginia	Fredericks	114797	30,393	37770.87
5	Virginia	Hanover	42110	116,423	3616.983
6	Virginia	Henrico	506023	342,775	14762.54
7	Virginia	Richmond	129928	237,257	5476.256
8	Virginia	Williamsbu	74190	15,861	46775.11
9	Virginia	Newport N	109799	183,230	5992.414
10	Virginia	Petersburg	60480	33,734	17928.5
11	Virginia	Norfolk	224375	231,013	9712.657

Fig. 3d: This .csv file of Dataset 2 includes rows for each county of interest for *Virginia Southern* in the second edition of the project, with columns for county name and state name, plus population of each county, total annual ridership of all Amtrak stations in each county, and normalized ridership data of ridership per 10,000 people.

In the first edition, I made a choropleth map in R and RStudio with my joined datasets for ridership per 10,000 people by county for *Mardi Gras Service*. Then, I used the Local Moran's I test on the choropleth map to analyze hotspots and clusters of values.

In the second edition, I expanded the analysis of *Mardi Gras Service* to include a log-transformed map of the counties as well as a z-score map. For *Borealis*, *Pacific Surfliner*, and *Virginia Southern*, I made simple choropleth maps, Local Moran's I maps, Global Moran's I scatterplots, log-transformed maps, and z-score maps.

Results

The choropleth map of ridership per 10,000 people in Figure 4 shows slightly higher than average ridership per 10,000 people in some of the counties with stations along the Mardi Gras Service route. However, there is one outlier for the entire dataset, Orleans Parish in Louisiana, which is where New Orleans is located. As expected, New Orleans skews the mean, and in examining the data without New Orleans, the highest value is less than 600 riders per 10,000 people. In that case, only one county along the route is significantly low for normalized ridership per 10,000 people, and the rest are either around the mean or not significantly lower, or even significantly higher than the mean.

Figure 4: Choropleth map - Mardi Gras Service (first edition)

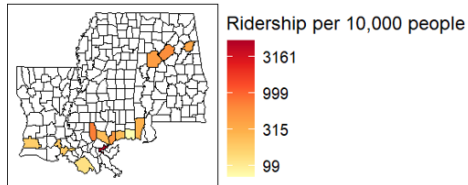


Fig. 4: Choropleth map made in R and RStudio that shows normalized ridership data by county for *Mardi Gras Service*, measured in ridership per 10,000 people.

In Figure 5, the map displays the results of the Local Moran's I hotspot test for significant hotspots of ridership per 10,000 people for each county. As expected, Orleans Parish, being the major outlier, is the only county considered a significant hotspot, with no other counties fitting such criteria.

Figure 5: Local Moran's I Hotspot Test - Mardi Gras Service (first edition)

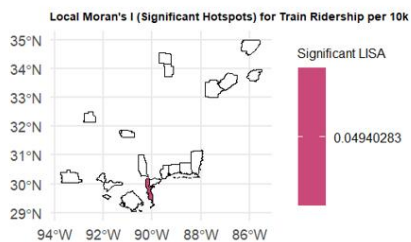


Fig. 5: Local Moran's I hotspot test for significant values for each county for *Mardi Gras Service*

After performing the hotspot test, I performed a cluster test with Local Moran's I. In Figure 6, the map shows clustering of high-high values where counties that are considered "clusters" are surrounded by high values. As expected, Orleans Parish, being an outlier, is the only county that is considered a "high-high" cluster and the rest are insignificant.

Figure 6: Local Moran's I cluster test – Mardi Gras Service (first edition)

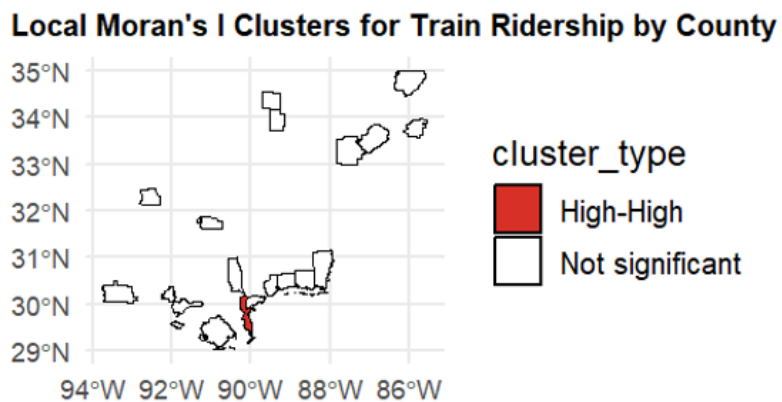


Fig. 6: Map that shows clustering of significant values by county for *Mardi Gras Service*

Since Orleans Parish has been a consistent outlier for the choropleth map, the hotspot test, and the cluster test, I decided to run the cluster test but exclude Orleans Parish. After running the Local Moran's I cluster test but excluding New Orleans, I produced the plot in Figure 7. This, however, produced a result that involved no significant clusters.

Figure 7: Local Moran's I cluster test, excluding Orleans – Mardi Gras Service (first edition)

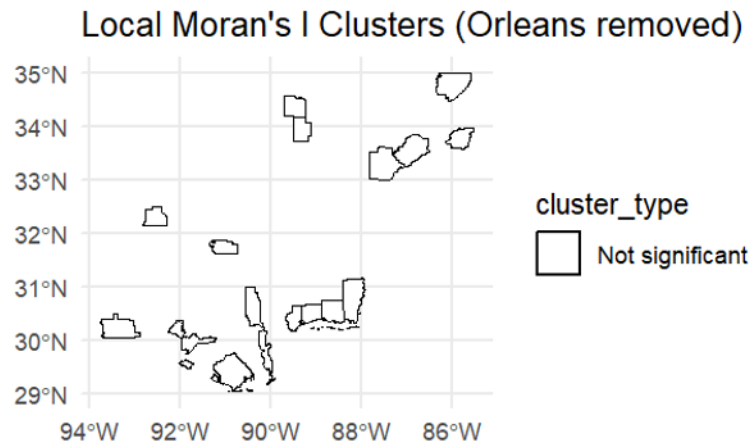


Fig. 7: A map showing the results of a Local Moran's I cluster test for *Mardi Gras Service* after removing Orleans Parish.

After revisiting this project, I decided to run the log-score for *Mardi Gras Service* since logarithms reduce skew and can show variation hidden by extreme values, which was necessary with extreme outliers like Orleans Parish. This resulted in a map with values of greater variance in Figure 8. The counties along the *Mardi Gras Service* route are similar to or higher than other counties with established service in value.

Figure 8: Log-transformed train ridership map – Mardi Gras Service (second edition)

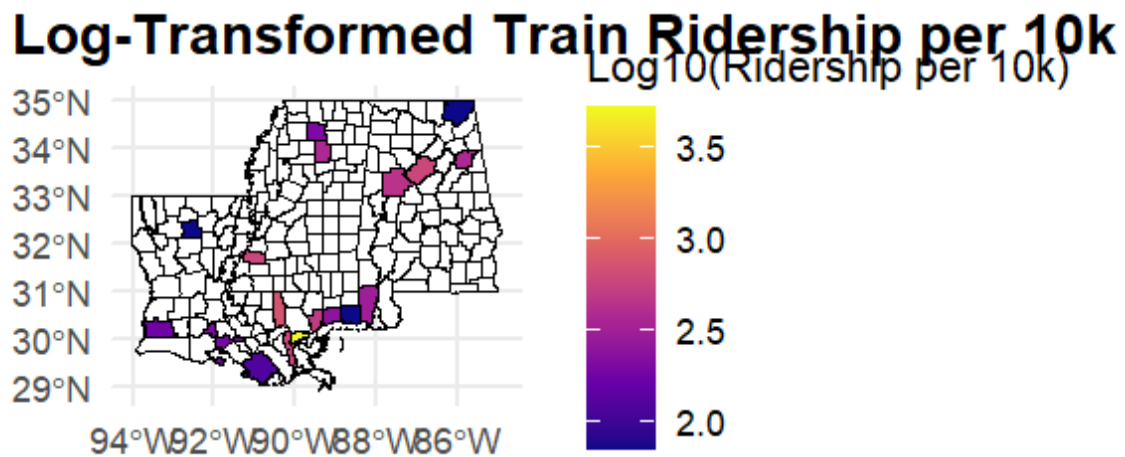


Fig. 8: A map showing the log-transformed values of ridership per 10,000 people by county for *Mardi Gras Service*

I also decided to run the z-score for *Mardi Gras Service*, which resulted in a map where Orleans was still an outlier and other counties were given a score of 0 in Figure 9.

Figure 9: Standardized (Z-score) map of ridership by county – Mardi Gras Service (second edition)

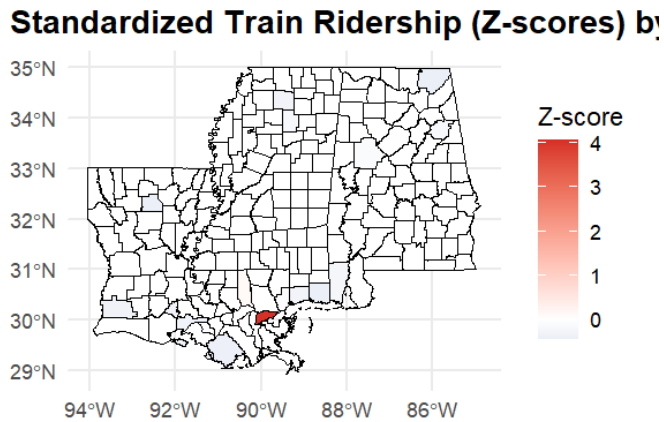


Fig. 9: A map showing the standardized values through z-scores for *Mardi Gras Service*

After finishing the *Mardi Gras Service* analysis, I began the *Borealis* analysis with a similar choropleth map of ridership per 10,000 people by county in Figure 10. While there are expected high values for Cook County, Illinois and Milwaukee County, Wisconsin, there is an unexpectedly low value for Ramsey County, Minnesota and an unexpectedly high value for Columbia County, Wisconsin. Possible explanations include Ramsey County, Minnesota being part of the Minneapolis-Saint Paul metropolitan area, and therefore having more options for transportation, including flights and buses. However, Columbia County, Wisconsin has 3 separate Amtrak stations, and Wisconsin Dells is one of those stations. The local tourism industry might explain the unusually high ridership of Columbia County.

Figure 10: Ridership per 10,000 people by county – *Borealis* (second edition)

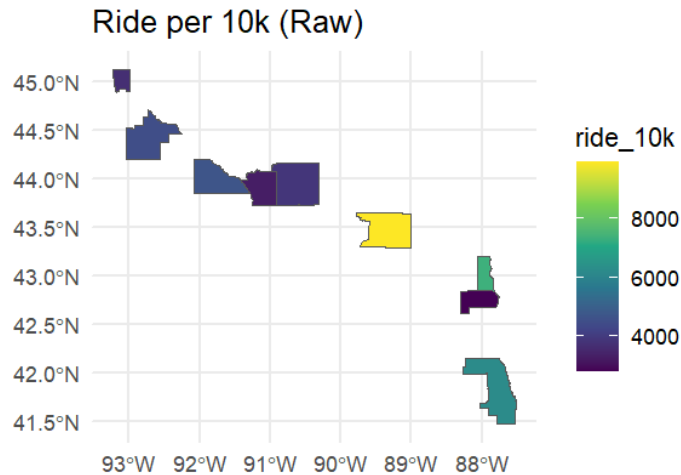


Fig. 10: A map showing ridership per 10,000 people by county for *Borealis*

I made a choropleth after running the Local Moran's I statistic for *Borealis*. The map in Figure 11 includes the values from each county. The lowest values are in Milwaukee County, Wisconsin and Racine County, Wisconsin. Milwaukee County had a higher normalized ridership value, while Racine County had a lower normalized ridership value. This is not surprising, as Milwaukee County includes a major hub in the city and a station at the airport as well. However, Racine County is a suburban county that is not a major hub, and many passengers in the northern suburbs of Milwaukee are likely to choose Milwaukee as their departure station. Monroe County has the highest value as it is next to a county with similarly low ridership, La Crosse County, which has a slightly lower value due to its proximity to Winona County, Minnesota. Winona County is higher in ridership compared to Monroe or La Crosse County.

Figure 11: Local Moran's I by county – *Borealis* (second edition)

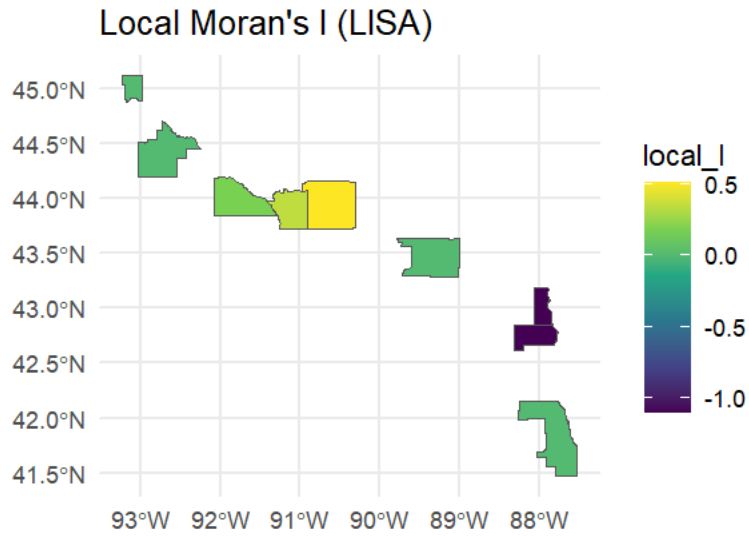


Fig. 11: A map showing the Local Moran's I test by county for *Borealis*

In the analysis for *Mardi Gras Service*, I did not include a Global Moran's I scatterplot, but I decided to include a scatterplot for *Borealis* in Figure 12 to see if there was clustering overall across this study area. The scatterplot produced a significant negative correlation. Therefore, there is not an overall trend of clustering. Rather, there is an overall trend of dispersion.

Figure 12: Global Moran's I scatterplot – *Borealis* (second edition)

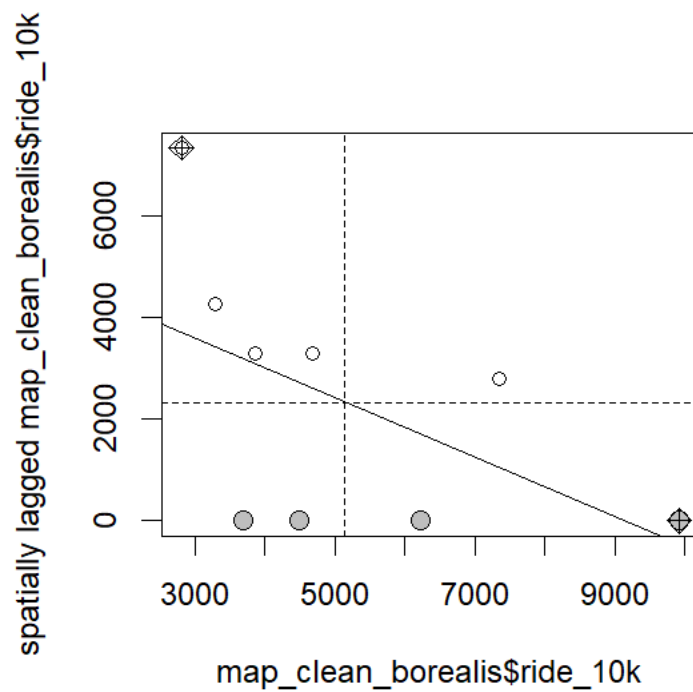


Fig. 12: Global Moran's I scatterplot for *Borealis*

I made a choropleth map of log-transformed normalized ridership values by county for *Borealis* in Figure 13. It has similar results to the simple choropleth in Figure 10.

Figure 13: Log-transformed map – *Borealis* (second edition)

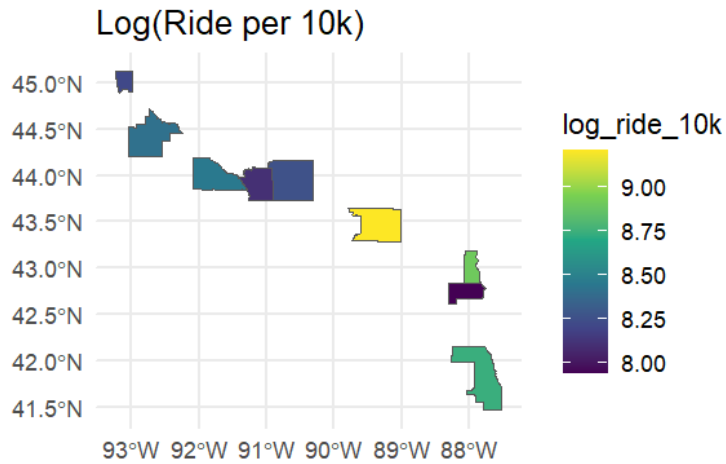
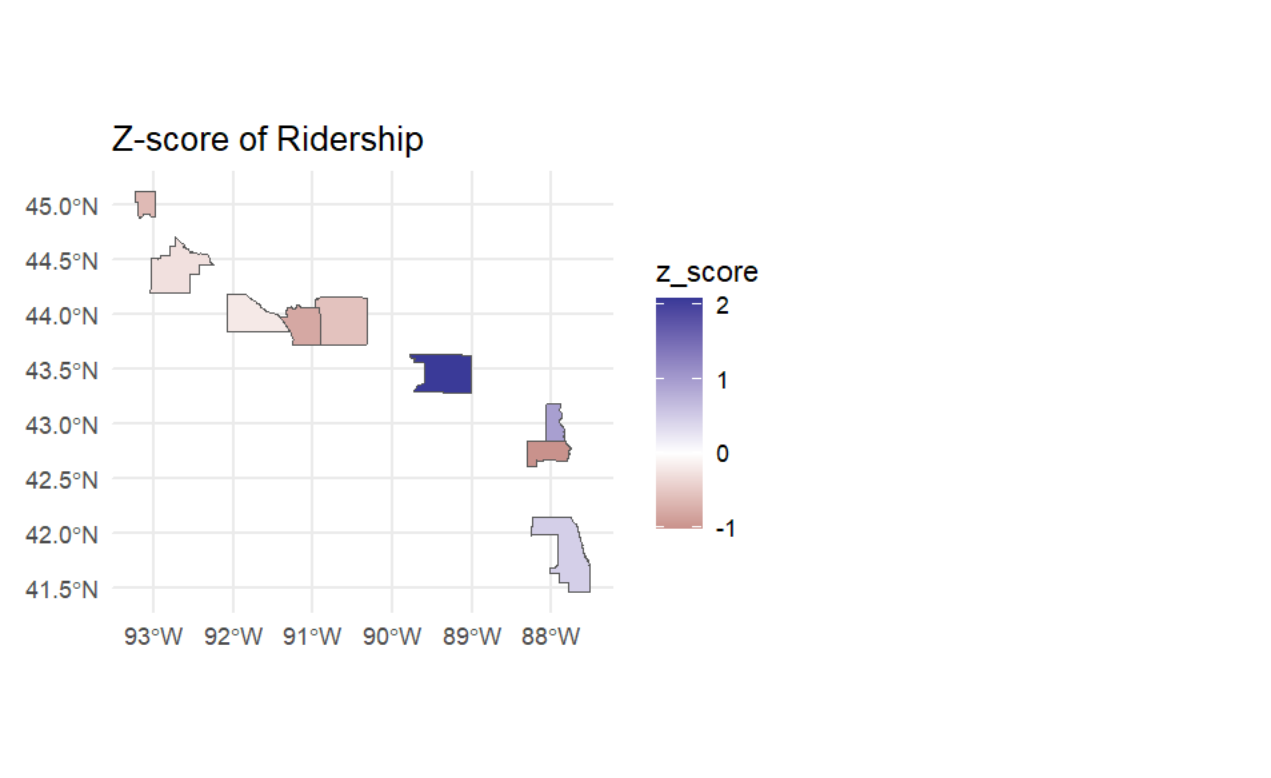


Fig. 13: A map showing the log-transformed values of ridership per 10,000 people by county for *Borealis*.

I ran the z-score for *Borealis*, which confirmed my assumptions that Columbia County is unusually high in ridership and Ramsey County is unusually low in ridership. Ramsey County has a negative z-score, and Columbia County has a positive z-score of 2. The results are in Figure 14.



For the next phase of the second edition, I made another normalized ridership by county map for *Pacific Surfliner* in figure 15. Surprisingly, Los Angeles County has the lowest ridership value per 10,000 people, while Orange County, San Diego County, and Ventura County all have lower-than-average normalized ridership numbers. Santa Barbara County has the highest normalized ridership values. In fact, Santa Barbara County has higher ridership than its entire population. This suggests *Borealis* has trends that are similar for *Pacific Surfliner*. Large urban hubs like Minneapolis-St. Paul and Los Angeles can be low in normalized ridership values and smaller and tourism-oriented communities can be high in normalized ridership values.

Figure 15: Ridership per 10,000 people by county – *Pacific Surfliner* (second edition)

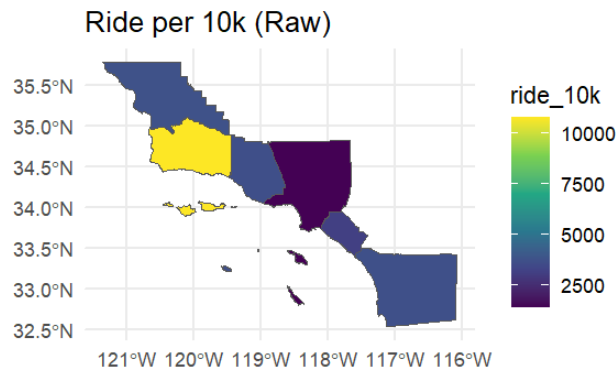


Fig. 15: A map showing ridership per 10,000 people by county for *Pacific Surfliner*

I made a choropleth after running the Local Moran's I statistic for *Pacific Surfliner*. The map in Figure 16 includes the values from each county. The lowest values are in San Luis Obispo County and Santa Barbara County. San Luis Obispo County has a lower normalized ridership value, while Santa Barbara County had a higher normalized ridership value. San Luis Obispo County shares a border with higher ridership Santa Barbara County, while Santa Barbara County shares a border on its other side with Ventura County, which is low in ridership. Los Angeles County has the highest value as it is next to counties with similarly low ridership, Orange County and Ventura County.

Figure 16: Local Moran's I – Pacific Surfliner (second edition)

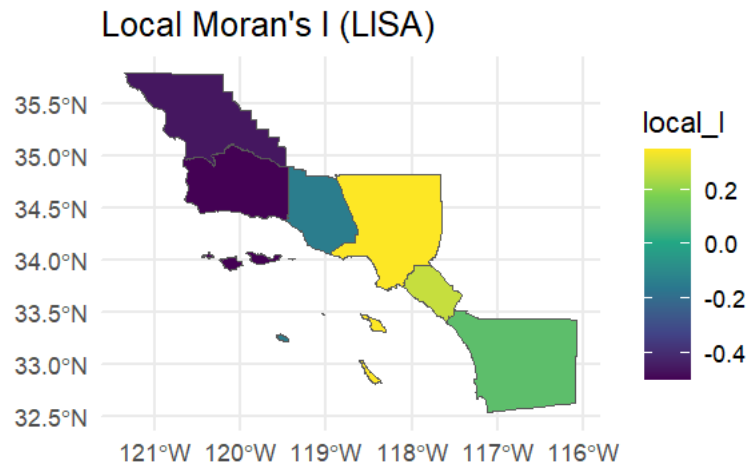


Fig. 16: A map showing the Local Moran's I test by county for *Pacific Surfliner*

I ran the Global Moran's I statistic and included a scatterplot for *Pacific Surfliner* in Figure 17 to see if there was clustering overall across this study area. The scatterplot produced a near-zero correlation. This implies that the ridership values are completely random without very many clusters.

Figure 17: Global Moran's I scatterplot – Pacific Surfliner (second edition)

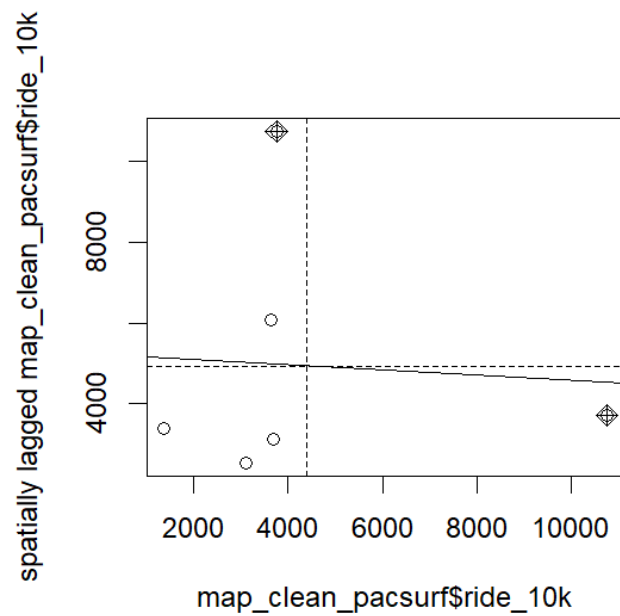


Fig. 17: Global Moran's I scatterplot for *Pacific Surfliner*.

I made a choropleth map of log-transformed normalized ridership values by county for *Pacific Surfliner* in Figure 18. It has similar results to the simple choropleth in Figure 15.

Figure 18: Log-transformed map – Pacific Surfliner (second edition)

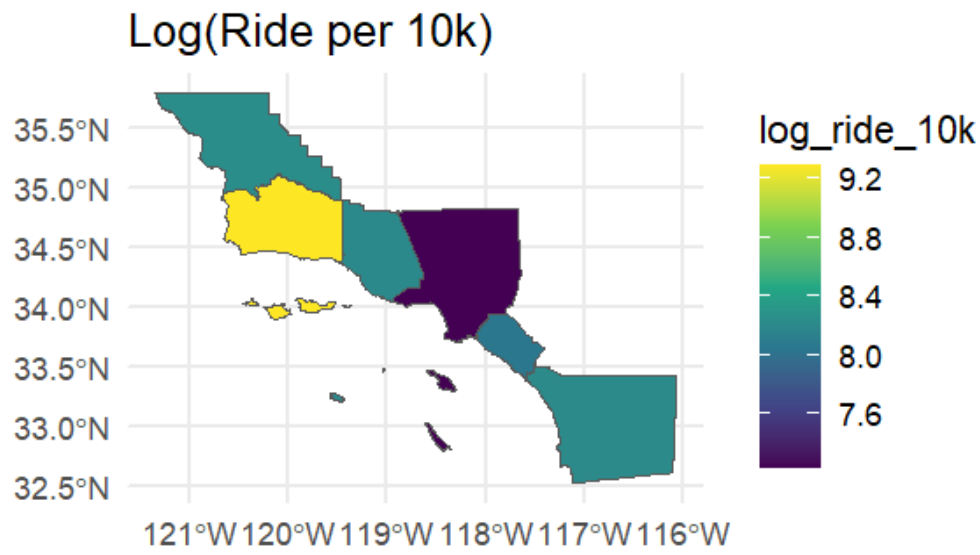


Fig. 18: A map showing the log-transformed values of ridership per 10,000 people by county for *Pacific Surfliner*.

I ran the z-score for *Pacific Surfliner*, which confirmed my assumptions that Santa Barbara County is unusually high in ridership and Los Angeles County is unusually low in ridership. Los Angeles County has a negative z-score of 1, and Santa Barbara County has a positive z-score of 2. The results are in Figure 19.

Figure 19: Standardized (z-score) values by county – *Pacific Surfliner* (second edition)

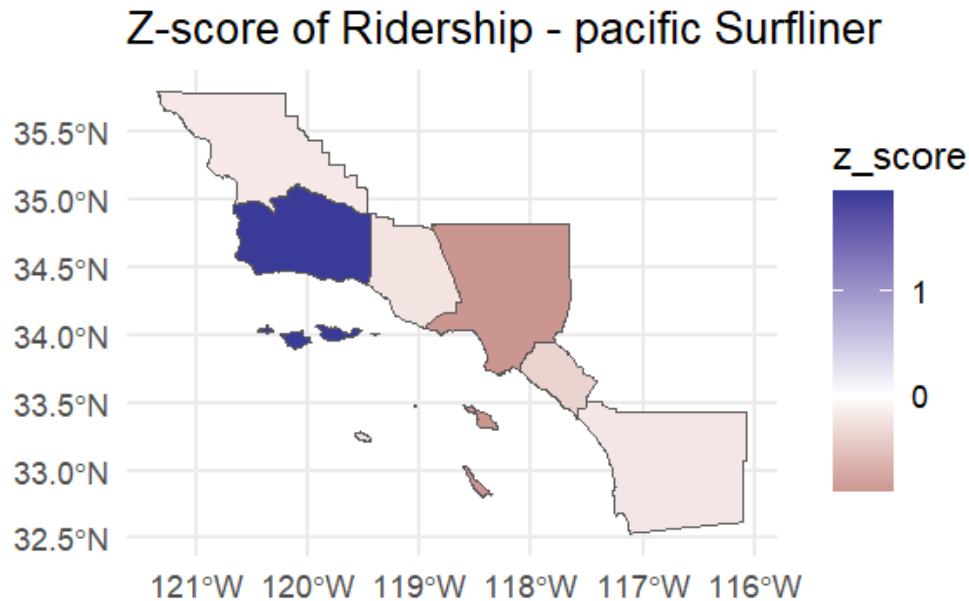


Fig. 19: A map showing standardized values (z-scores) by county for *Pacific Surfliner*

For the final phase of the second edition, I began with another normalized ridership by county map for *Virginia Southern* in Figures 20a and 20b. I observed a trend like *Pacific Surfliner*, and to a lesser extent, *Borealis*. Larger cities and suburban counties like Richmond, Norfolk, and Prince William County had lower ridership, while locations like Williamsburg that are smaller and tourism-dependent had high ridership.

Figure 20a: Ridership per 10,000 people by county – Virginia Southern (second edition)

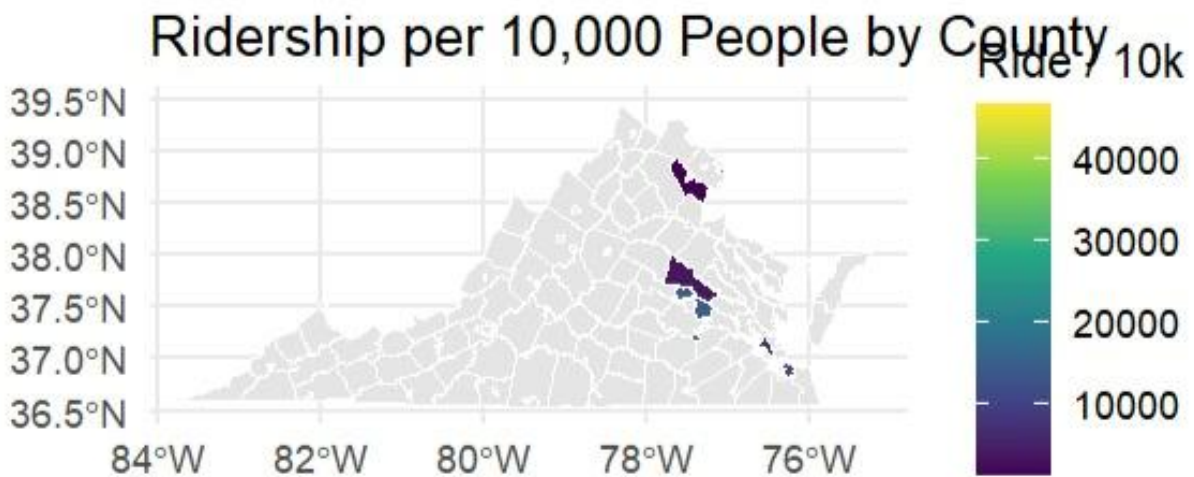


Fig. 20a: A map showing ridership per 10,000 people by county for *Virginia Southern*

Figure 20b: Ridership per 10,000 people by county- Virginia Southern; zoomed in (second edition)

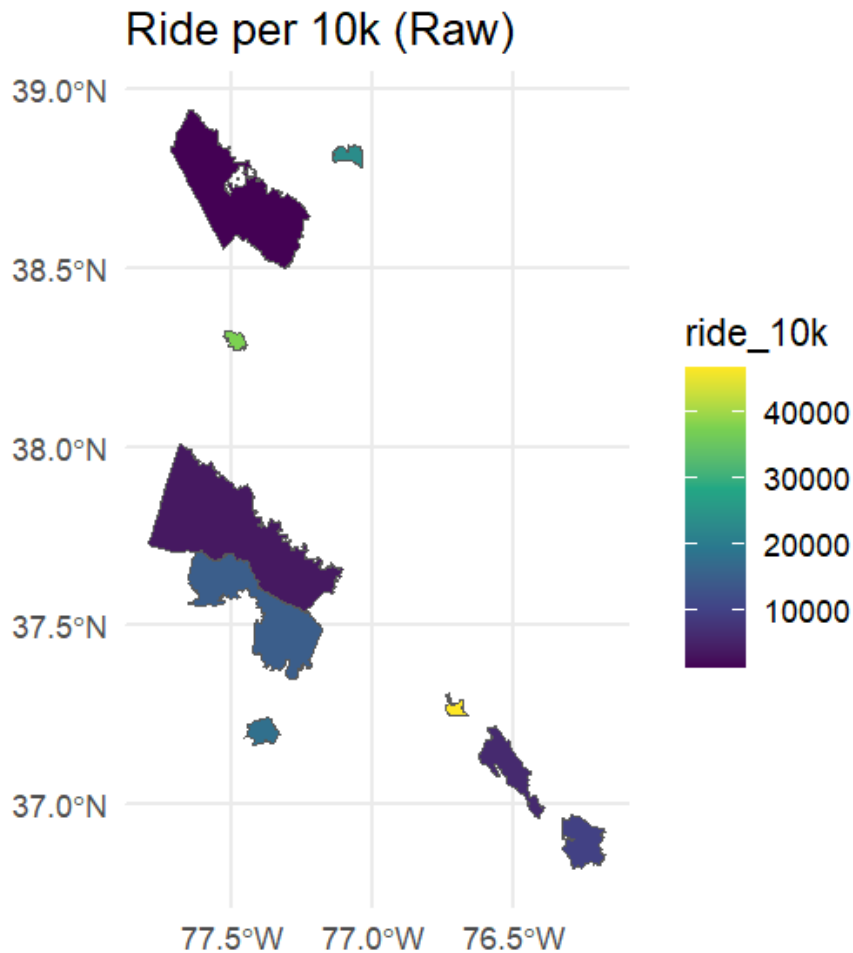


Fig. 20b: Map similar to 20a but zoomed in to see smaller counties and county-level divisions.

I made a choropleth after running the Local Moran's I statistic for *Virginia Southern*. The map in Figure 21 includes the values from each county. The lowest values are in all counties outside of the Richmond area. Norfolk and Newport News have a lower normalized ridership value, while Williamsburg has a higher value. Similarly, Alexandria has a higher-than-average value while Prince William County has a lower-than-average value. However, the counties that border each other in the Richmond area are both lower-than-average in ridership values.

Figure 21: Local Moran's I by county – *Virginia Southern* (second edition)

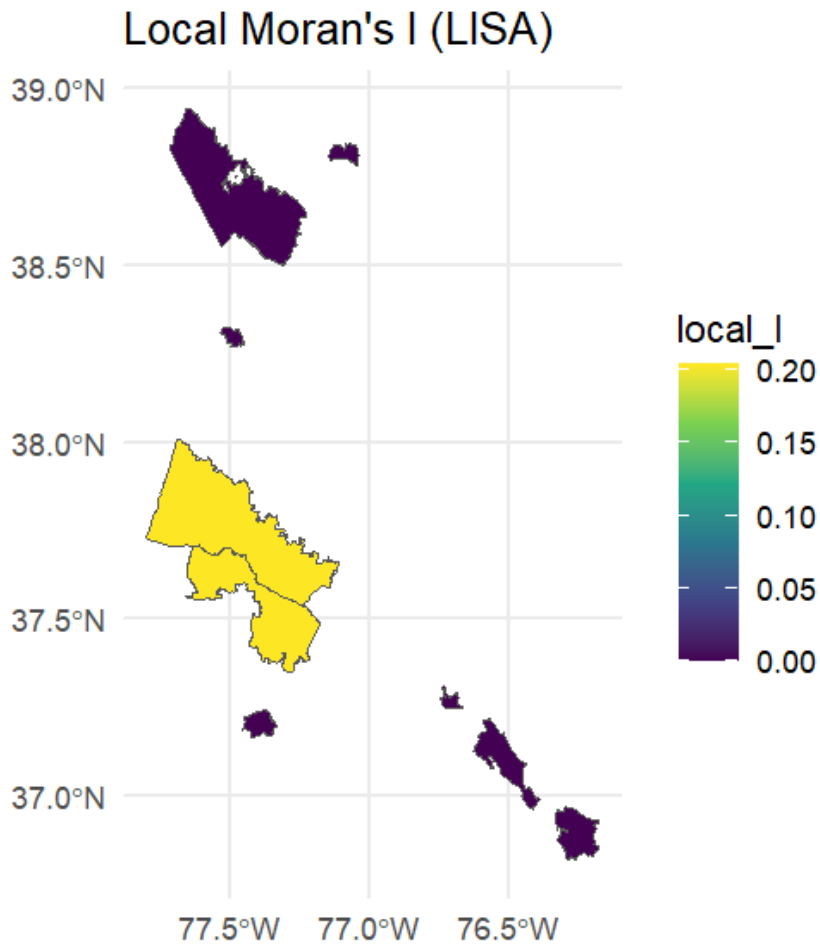


Fig. 21: A map showing the Local Moran's I test by county for *Virginia Southern*

I ran the Global Moran's I statistic and included a scatterplot for *Virginia Southern* in Figure 22 to see if there was clustering overall across this study area. The scatterplot produced a slightly significant negative correlation. Therefore, there is not an overall trend of clustering. Rather, there is a slight trend of dispersion.

Figure 22: Global Moran's I scatterplot – *Virginia Southern* (second edition)

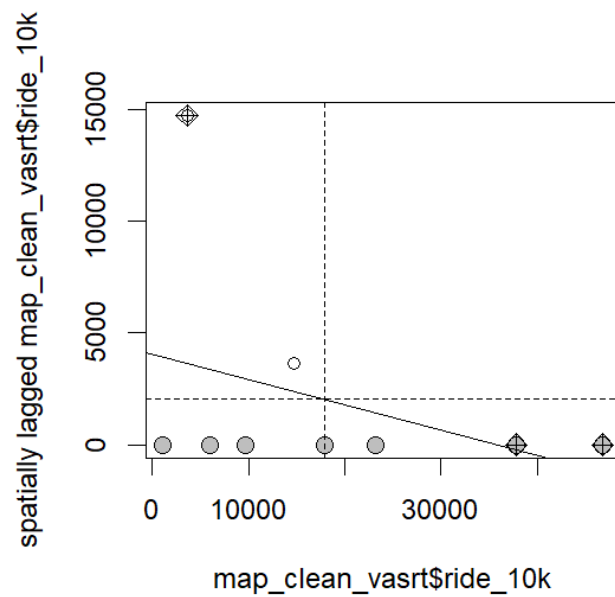


Fig. 22: Global Moran's I scatterplot for *Virginia Southern*

I made a choropleth map of log-transformed normalized ridership values by county for *Virginia Southern* in Figure 23. It has similar results to the simple choropleths in Figures 20a and 20b.

Figure 23: Log-transformed map by county – Virginia Southern (second edition)

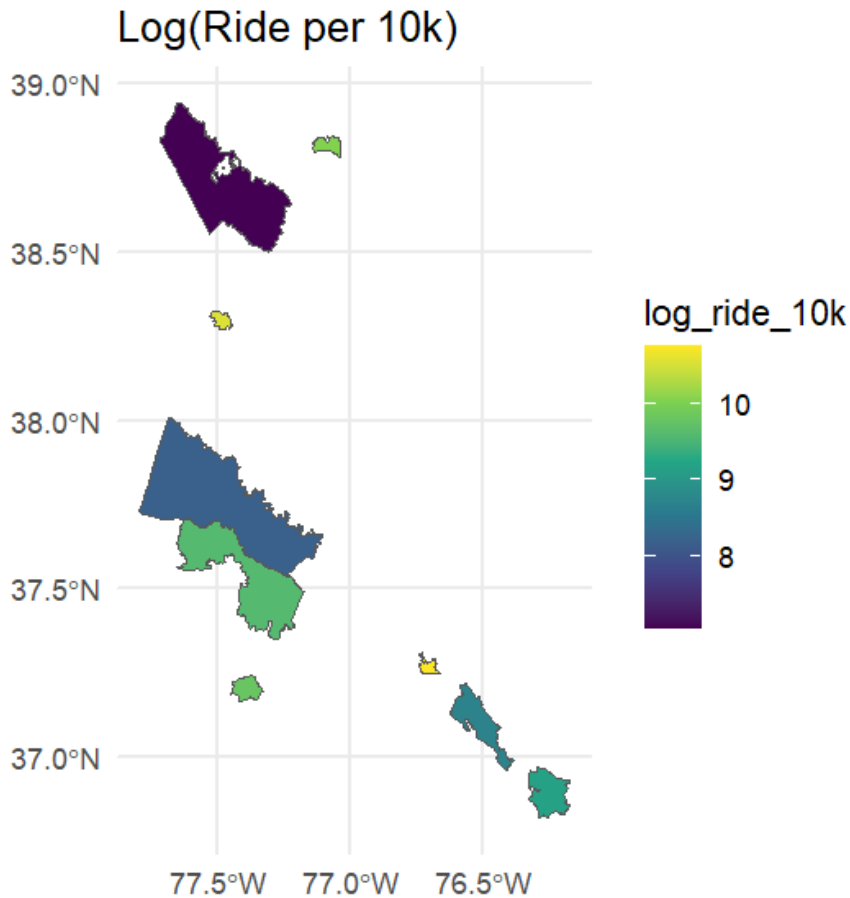


Fig. 23: A map showing the log-transformed values of ridership per 10,000 people by county for *Virginia Southern*

I ran the z-score for *Virginia Southern*, which confirmed my assumptions that Williamsburg is unusually high in ridership and Prince William County is unusually low in ridership. Prince William County has a negative z-score of 1, and Williamsburg has a positive z-score of 1. The results are in Figure 19.

Figure 24: Standardized (z-score) values by county – *Virginia Southern* (second edition)

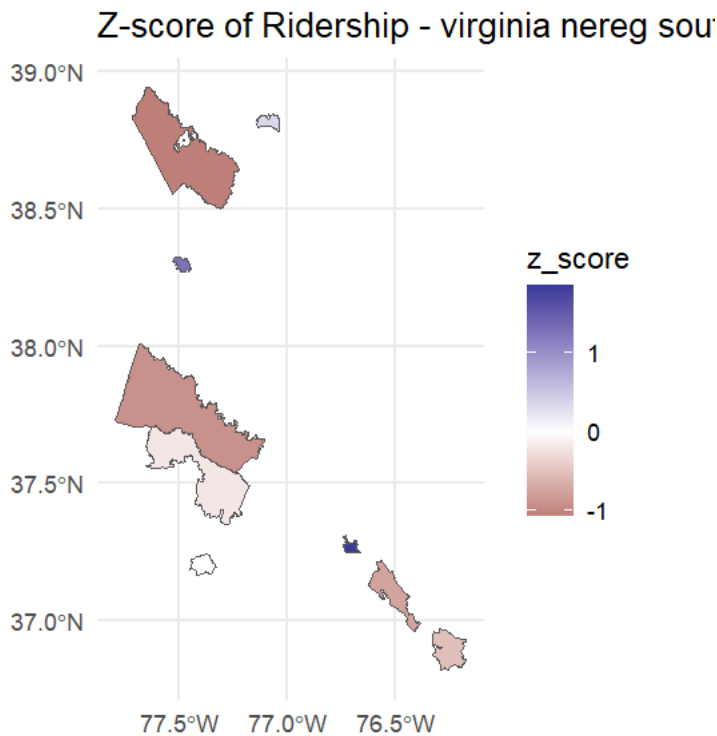


Fig. 24: A map showing standardized values (z-scores) by county for *Virginia Southern*

Discussion

By finding ridership and population data through primary sources with Dataset 1 and Dataset 2, I was able to easily integrate the basemaps in R and RStudio for states and counties by joining the tables. In doing so, I created a choropleth map with normalized data for ridership per 10,000 people by county in the first edition of this project. Orleans was an outlier, and this led me to perform Local Moran's I hotspot and cluster tests. In both tests, when every county of interest was included, Orleans was the only significant hotspot and cluster. Therefore, I ran the cluster test while removing Orleans from the dataset. However, none of the counties were significant clusters in a test that removed Orleans.

These results from the Local Moran's I tests suggest that there is not significantly high ridership in the counties along the *Mardi Gras Service* route, but there is also not significantly low ridership along the same route. This can be interpreted to confirm the theory of induced demand, where new demand is created by the expansion of rail access. This is an example of demand elasticity. This confirms the previous public statements made by public agencies, where such agencies emphasized the service performing better with ridership than initially expected. This could be a result of initially assuming that demand for passenger rail is inelastic rather than elastic. This could be a reason to modify quantitative analysis modeling for estimating passenger rail demand to factor in the concept of induced demand and demand elasticity. Such a modification for quantitative analysis modeling would change how governments, policymakers, transportation economists, and railroad planners estimate demand and make decisions regarding rail expansion.

However, this method does have its limitations. As previously stated, while there were no significantly low clusters, there were also not any significantly high clusters. This suggests that the Local Moran's I test has many limitations with data that is more discrete and spatially separated.

If the first edition of the project had been afforded infinite resources and time, more quantitative spatial analysis techniques could have been utilized to find one that might work better than the Local Moran's I tests. For the second edition, I was able to run logarithmic scores and z-scores for the *Mardi Gras Service*. I then used the simple choropleth method with normalized ridership numbers, the Local Moran's I test, the Global Moran's I scatterplot, logarithmic scoring, and z-scoring for *Borealis*, *Pacific Surfliner*, and *Virginia Southern*. These study areas showed a more

common trend of rural outliers for ridership, which further reinforces the theory of induced demand.

By further analyzing spatial data regarding train ridership and exploring different quantitative spatial analysis techniques, we can explore and overcome the limitations of our current models that are used in our analysis to predict passenger rail demand. This will allow policymakers, transportation economists, governments, and railroad planners to make better-informed decisions that factor in different variables.

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